

# I 5. The Computational Theory of Mind

Start with motivations for CTM- quotes from Haugeland.

Reorient towards: is CTM possible? Focus on problem issues: emotion, consciousness, intelligence.

## CTM Defined

**CTM v.1:** “The central tenet of the theory is that a mind is a computer.” (McGlaughlin)

“Three famous philosophical dilemmas stood in the way [of a materialist theory of mind]: (i) the metaphysical problem of mind interacting with matter; (ii) the theoretical problem of explaining the relevance of meanings, without appealing to a question-begging homunculus; and (iii) the methodological issue over the empirical testability (and, hence, respectability) of "mentalist" explanations. The computational idea can be seen as slicing through all three dilemmas at a stroke; and this is what gives it, I think, the bulk of its tremendous gut-level appeal. (Haugeland 1981)

**Issue 1:** is this claim a metaphor, a methodology, or a literal identification?

**Comment:** In this class, it is meant literally: the mind is a computer just the way the heart is a pump. That is its actual biological function. But this is a philosophical conjecture. In cognitive science, computationalism is treated more like a methodological assumption than a world-view.

**Issue 2:** how could it be a literal claim, given that our brains are obviously *not* “computers” like laptops or phones. Isn’t the target claim clearly metaphorical?

**Comment:** this is just a verbal confusion. The technical concept of a “computer” is that of a system whose function is to perform cognitive tasks (rational transitions) that relate cognitive contents by mechanically manipulating symbols which function to represent those contents—i.e. an information processor. This definition is neutral about the physical composition of the system or its history. It’s in this sense that we mind is literally a computer.

**Issue 3:** is the claim that the *mind* is a computer, or that the *brain* is a computer?

**Comment:** Really, this is a claim about the brain, since computers are *embodied* systems. Instead, the mind is better thought of as what the brain *does*. Perhaps we should say, the mind *emerges* from a computer (the brain).

**Issue 4:** is the idea that the mind is a *computer* or that it’s a *computer program*?

**Comment:** if the brain is a *computer*, perhaps the mind is a *computer program*? This might be half right, in the sense that a computer executes (or performs) a computer program. But as we’ll see, the concept of a “program” refers specifically to instructions that are distinct from the hardware they are stored in. But much of the mind emerges from biological hardware. The more general concept is that of an *algorithm*.

**Issue 5:** is this a claim about all aspects of the mind, including consciousness and emotion?

**Comment:** While consciousness may be a computational state of the brain, there are few computationalists (even methodological computationalists) who believe that computation can *explain* phenomenology. In the end, computationalism is really about the cognitive mind. Emotion remains an unsettled case.

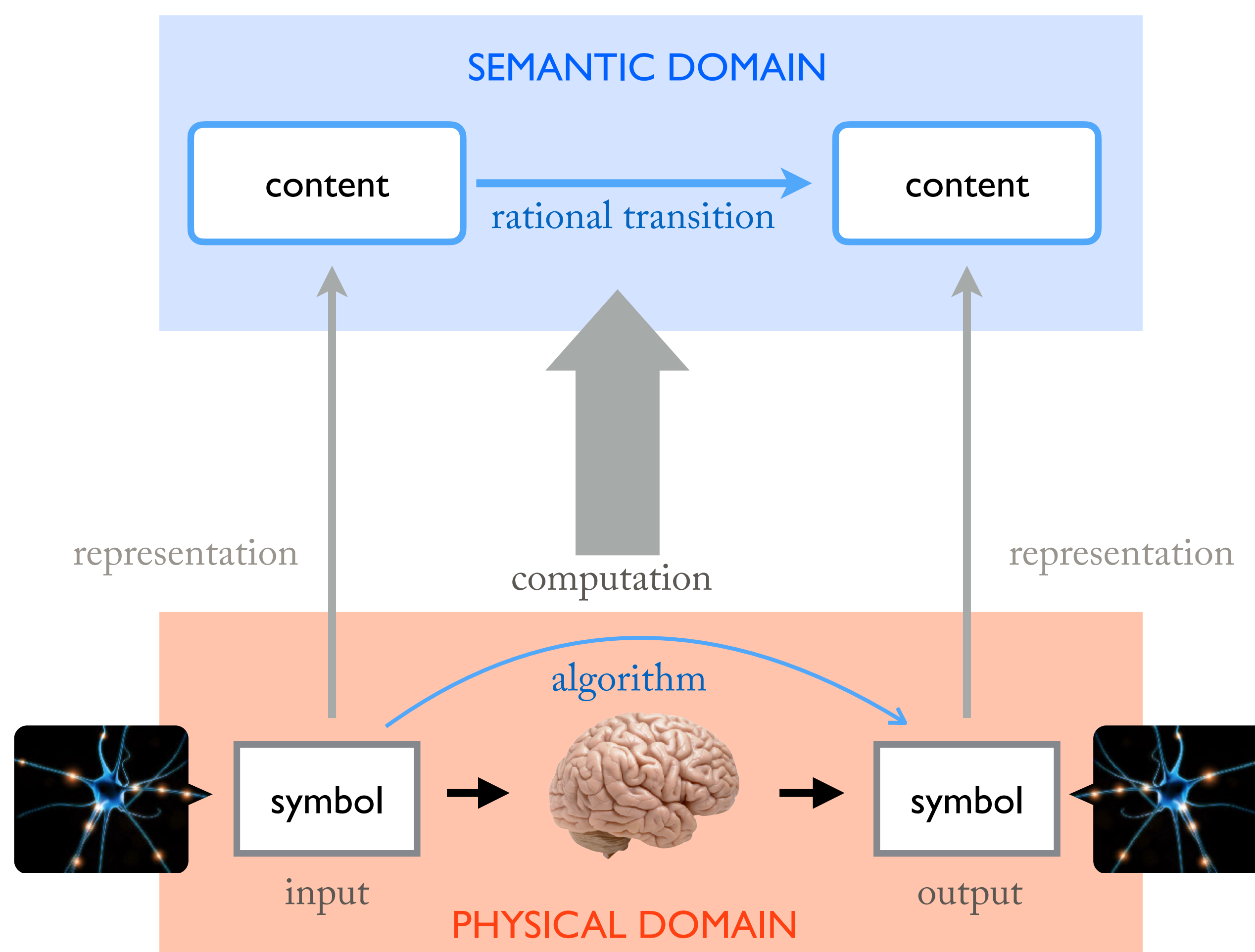
**Issue 6:** what is meant by “the mind” anyway, given that we have many different kinds of mental states, which constantly changing over time?

**Comment:** the concept of *mind* is not particularly clear one. It’s probably better to talk about mental *capacities*, mental *states*, and mental *processes*. (The mind, roughly, is the unity of these elements.) The idea is that mental states are computational states of the brain, and mental processes are computational processes of the brain.

**Issue 7:** how could the brain be a computer, since no one designed a brain, or assigned it an interpretive code?

**Comment:** the brain wasn’t designed, but it does have an overall *biological* function, and its parts have many specific biological functions. States of the brain function to carry information about the world (to represent); and structures in the brain function to process these states (or, compute).

**Computationalism v. 2:** the brain is a natural computer, or information processor. The biological function of the brain is to perform a range of cognitive functions by carrying out mechanical transformations on physical symbols which compute those functions. Cognitive states are computational states of the brain. Cognitive processes are computational processes of the brain.



**Computational Theory of Mind:** the brain computes cognitive functions by manipulating symbolic representations in the brain according to biologically grounded algorithms.